

# Feel What You See: A Novel Sensory Interface Linking Visual Heat Cues and Instant Thermal Feedback

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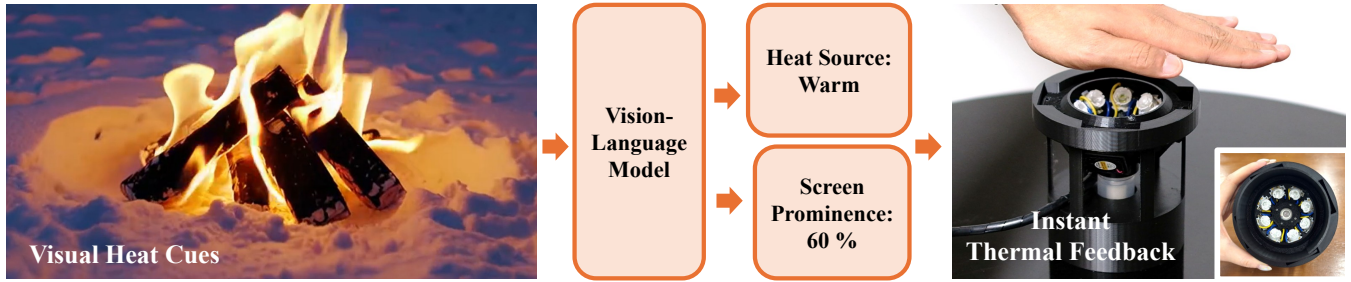


Figure 1: "Feel What You See": The system analyzes visual heat cues and delivers immediate temperature sensation.

## Abstract

We present a novel sensory interface that allows users to truly feel what they see by transforming visually implied heat cues into physical temperature sensations. Using a vision-language model, the system semantically analyzes video content to detect heat-related elements—such as fire, ice, or sunlight—and determines the appropriate thermal response based on both type and screen prominence. The core of this experience is a custom-designed, non-contact thermal device combining two physical mechanisms: an ethanol-based aerosol module for cooling, and a visible-spectrum radiation module for heating. The system dynamically adjusts the type and intensity of stimulation to reflect the semantic properties of each visual scene. This prototype demonstrates a new form of semantic-driven multisensory interaction, extending the expressive power of visual media. It opens new possibilities in immersive storytelling, sensory accessibility, and emotion-enhanced interfaces—where the ambient temperature of a scene can be directly felt, not just seen.

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## CCS Concepts

• **Human-centered computing** → **User interface design**; *Interaction devices*; • **Hardware** → *Haptic devices*.

## Keywords

thermal feedback, vision-language model, sensory interface

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## 1 Introduction

In immersive media and virtual environments, vision dominates how users perceive context and narrative, while thermal sensation—one of the most fundamental and emotionally resonant modalities of human perception [Jones and Ho 2008]—is rarely considered. Incorporating thermal feedback can significantly enhance realism and immersion in applications such as storytelling, simulation, and affective interfaces.

Existing systems often rely on predefined cues or contact-based Peltier elements [Lee et al. 2020; Peiris et al. 2017; Wang et al. 2024b,a], which are slow and unsuitable for dynamic scenes [Xu et al. 2024]. Recent efforts using large language models to infer thermal impressions from text [Nam et al. 2024] represent progress

toward semantic-driven thermal rendering, but they overlook visual cues such as fire, steam, or sunlight, as well as their spatial prominence.

To bridge this gap, we present Feel What You See, a semantic-driven interface that maps visual heat cues to real-time, non-contact feedback. Leveraging a vision-language model, our system identifies heat-related elements in video frames and estimates their spatial coverage. A custom device then delivers multi-level heating and cooling, allowing users to physically “feel what they see.”

## 2 Concept - Feel What You See

Our perception of temperature is strongly shaped by visual cues such as fire, ice, sunlight, or steam. Their size and position within a scene often imply distance and intensity, shaping how we anticipate thermal experiences. Building on this connection, we introduce the concept of Feel What You See—an interaction paradigm that transforms semantic heat cues in visual media into physical sensations. By combining vision-language analysis with a non-contact thermal feedback system, this approach enables users to experience temperature changes in real time, reinforcing presence, emotional resonance, and narrative immersion.

## 3 Implementation

### 3.1 Hardware

As shown in Figure 2, the device integrates two modules for heating and cooling. Heating is provided by eight high-intensity 5W visible-spectrum LEDs, while cooling is achieved with an ultrasonic atomizer dispersing a 20 % ethanol solution. Users describe the sensations as ranging from fire-like warmth to ice-like coolness, with rapid switching between hot and cold that Peltier systems cannot achieve. Compared to our earlier prototype, MoHeat[Xu et al. 2024], which required vortex airflow and an external compressor, this mist-based design offers a more portable and responsive solution, with outputs controlled via serial communication for real-time adjustment.

### 3.2 Software

As shown in Figure 3, the software pipeline connects visual semantics to thermal feedback through three stages: content analysis, result analysis, and interface rendering. Input video frames are first processed by Qwen2.5-VL-7B-Instruct [Team 2024], which identifies heat-related elements such as fire, ice, steam, or sunlight and their screen coverage. Unlike low-level methods based on color or brightness, VLMs provide higher-level semantic abstraction, enabling the system to recognize the types of thermal cues in context. A mapping algorithm then classifies these elements into thermal categories and assigns stimulus intensity (e.g., a large fire region triggers stronger heating than a small patch of sunlight). The current system mainly supports single hot–cold mappings, while more complex scenarios—such as mixed stimuli (“fire in the snow”) or proportional effects (a small but intense source vs. a large mild one)—remain open for future user studies. Finally, results are integrated with the video stream through a PyQt5 GUI [Limited 2024], which communicates with the device via USB serial commands specifying mode, duration, and power. The system supports both frame- and scene-level processing, includes a calibration interface for adjusting thresholds

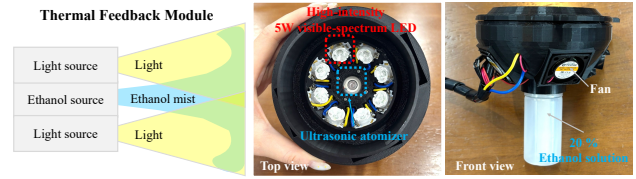


Figure 2: Design for Hardware.

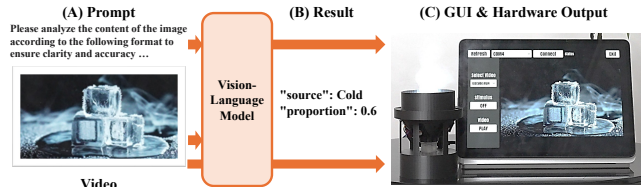


Figure 3: Flowchart for software.

and timing, and all GUI components and VLM code are publicly available at: [https://github.com/ChenKaiXuSan/Temp\\_Feedback](https://github.com/ChenKaiXuSan/Temp_Feedback).

## 4 User Experience

At SIGGRAPH Asia, attendees will sit in front of a display and watch short curated clips featuring fire, ice, steam, and sunlight. As the scenes shift—for example, from a blazing fire to a snowy landscape—they will directly feel the corresponding change in temperature through the system’s non-contact feedback. This demonstration emphasizes the audience’s sensory experience, showing how visual content can be extended with thermal cues to deepen immersion and emotional engagement, with potential applications in storytelling, education, simulation, and affective computing.

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